XRPL Validator — Daily Engineering Report · 2026-09-13

Period: 2026-09-13 00:00 to 24:00 US Eastern (2026-09-13 04:00 to 2026-09-14 04:00 UTC); ledgers #106,949,450 to #106,971,913. Every count from the ledger and the validator's logs stops at #106,971,913. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 6 engine receipts recorded (0 fixed, 0 explained, 6 without a recorded fix).
- **Changes:** 3 deployments (cycles 124, 126, 127), 9 commits merged into main, 11 engine findings fixed (F267–F277).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,949,450–#106,971,913), by step:

Step	Ledgers with a receipt
Step 1 · native engine	6
Step 2 · records	0

Step	Ledgers with a receipt
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,951,814	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: EC8505E9, EF5A72D8, 9F52FE87 (deleted)	no disposition recorded
R-02	—	#106,952,542	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-03	—	#106,953,156	Step 1 · native engine	13 change(s) of ours the network did not make	no disposition recorded
R-04	—	#106,953,944	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-05	—	#106,954,035	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-06	—	#106,954,058	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-13-01, INC-01 is INC-2026-09-13-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
124	`e094350`	05:42:36 UTC	n/a	n/a	411 matched, 0 mismatched	#3 from 05:42 UTC: ended early: superseded by the next deployment (2026-09-13 15:07 UTC)
126	`617c445`	15:07:36 UTC	n/a	n/a	416 matched, 0 mismatched	#4 from 15:07 UTC: ended early: superseded by the next deployment (2026-09-13 17:09 UTC)
127	`88794e2`	17:09:31 UTC	n/a	n/a	419 matched, 0 mismatched	#5 from 17:09 UTC: ended early: superseded by the next deployment (2026-09-14 16:00 UTC)

Cycle 124 delivered: `b0ec4bd` chore(watch_engine): ter-miss breakdown counts this run's receipts; upstream panel; no LAN IP; `357975b` fix(xrpl-ledger): F266 — a token escrow finished by its destination creates the missing trust line; `e094350` fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee (tecINSUFFICIENT_PAYMENT)

Cycle 126 delivered: `123e9a7` fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly; `1eb4a78` fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built; `9f9f231` fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free; `c748500` fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing; `a7c12f5` fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge; `617c445` fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty

Cycle 127 delivered: `d48c66a` fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold; `88794e2` fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
04:28 UTC	xrpl-validator	`e094350`	fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee (tecINSUFFICIENT_PAYMENT)
12:43 UTC	xrpl-validator	`123e9a7`	fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly
12:43 UTC	xrpl-validator	`9f9f231`	fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free
12:43 UTC	xrpl-validator	`1eb4a78`	fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built
13:57 UTC	xrpl-validator	`c748500`	fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing
13:57 UTC	xrpl-validator	`a7c12f5`	fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge
13:57 UTC	xrpl-validator	`617c445`	fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty
15:56 UTC	xrpl-validator	`d48c66a`	fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold
15:56 UTC	xrpl-validator	`88794e2`	fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial

Findings fixed

- **fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee (tecINSUFFICIENT_PAYMENT)** (`e094350` , xrpl-validator) rippled 3.3.0 NFTokenAcceptOffer::preclaim, brokered mode (:95-142): same token, same asset, no owner loop, so.Amount > bo.Amount -> tecINSUFFICIENT_PAYMENT; then, with an explicit NFTokenBrokerFee, the fee must be the buy asset, below the buy amount, and the ask must still be covered after it. Our preclaim had none of the amount checks.

- **fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly** (`123e9a7`, xrpl-validator) rippled sizes an OUT-limited partial fill of a book maker with `limitStepOut` → `TOffer::limitOut` → `Quality::ceilOutStrict` (Offer.h:216- 221): a true ceiling of the exact product to whole drops. Our bridge's `b_unprice` used `mul_round16_up`, the lossy legacy `mulRound`, which discards everything below the 17th digit before its carry.
- **fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free** (`9f9f231`, xrpl-validator) `BookOfferCrossingStep::adjustQualityWithFees` (BookStep.cpp:512-540) returns the offer quality untouched for OfferType::Clob — "Offer x-ing does not charge a transfer fee … single path AMM's offer quality is not constant" — so only a single-path POOL leg carries trIn into the strand's QualityFunction; a book tip enters fee-free, while the limitQuality the function is solved against stays the fee-inflated one. F224 …
- **fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built** (`1eb4a78`, xrpl-validator) rippled's strand builder runs `checkFreeze(view, src, dst)` on the delivery step with dst = the taker (StepChecks.h; DirectStepI::check for an IOU, XRPEndpointStep::check with src = xrpAccount for XRP). It returns terNO_LINE when the taker's OWN AccountRoot carries lsfGlobalFreeze, or, for an IOU delivery, when the taker's line to the issuer carries the taker-side freeze bit or either deep-freeze bit. `toStrand` …
- **fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing** (`c748500`, xrpl-validator) `size_book_hop` measures a hop by granting 1e6 / 1e12 / 1e18 of the in-asset and reading what came out. A rung that yields NOTHING is not "bound by liquidity", it is below the hop's quantum: a pool priced at ~2.8e6 FACE per drop turns 1e6 FACE into 0.36 drops, which floors to zero. `consumed == 0 < granted` then read as a liquidity bound, the hop reported 0, the hop before it got the UNBOUNDED cap and consumed its …
- **fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge** (`a7c12f5`, xrpl-validator) When the whole drop the OUT side ceiled to costs more than the taker has left, rippled's forward pass is in-limited and leg A yields `floor(in / rate)` drops (F93's `limitStepIn` → `ceilInStrict(…, roundUp = false)`; a pool's `swapAssetIn` output rounds down the same way). We clamped `gets_in` to the remainder while keeping the ceiled drop — buying a drop the taker could not pay for.
- **fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty** (`617c445`, xrpl-validator) `AMMLiquidity::getOffer` answers nullopt once `ammContext.maxItersReached()` (kMaxIterations = 30); the counter advances once per flow iteration whose winning pass used the pool (`setAMMUsed` → `update`). The offer walk IS the flow, so every slice it takes is one such iteration. Finding 107 gated only `discover`, which the walk calls once — the 31st slice and every one after it went through, and the single-path turn …
- **fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold** (`d48c66a`, xrpl-validator) rippled's `remainingIn = *sendMax - sum(savedIns)` (StrandFlow.h:745) is the sixteen-digit sorted-multiset fold of the ITERATIONS' ins — one entry per book level or pool slice — and a FillOrKill sell is judged on that (:829-841: remainingIn must be zero or the flow returns tecPATH_PARTIAL and the offer is tecKILLED). The direct walk folded per FILL for Finding 236's rested amount, skipped that fold when its running …
- **fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial** (`88794e2`, xrpl-validator) rippled's FlowOfferStream deletes an offer whose owner's funds are zero NOW but were not in the pristine view — "became unfunded" — in the strand's sandbox; the removal persists only if that strand wins the iteration and its sandbox is applied. A rejected trial ("Path rejected by limitQuality") is

discarded, removal included. Only a "found unfunded", expired or domain removal is `permRmOffer`ed for good …

- **fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa** (`03b96c8`, xrpl-validator) `divide` hands `muldiv(numVal, 1e17, denVal) + 5` to `STAmount(asset, uint64, exponent, negative)`, whose `canonicalize()` builds `Number(negative, value_, offset_)` — and Number's mantissa is a signed 64-bit. A quotient at or above 2^63 wraps negative; the magnitude that survives is 2^64 − v, and `getRate` files it with the sign dropped (STAmount.cpp canonicalize / getRate, rippled 3.3.0).
- **fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent** (`5d804cc`, xrpl-validator) rippled's FlowOfferStream splits the reduced-quality (tiny) removal the way it splits the unfunded one (OfferStream.cpp:308-332): "Removing tiny offer due to reduced quality" is `permRmOffer` when the owner's funds are what the pristine view held, "Removing tiny offer that BECAME tiny due to reduced quality" lives in the strand's sandbox otherwise — discarded with a rejected trial or a killed transaction. Four of …

Independence program

No independence record for UTC day 2026-09-13; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 94.

Open items

- Receipt R-01 (#106,951,814): open, under investigation.
- Receipt R-02 (#106,952,542): open, under investigation.
- Receipt R-03 (#106,953,156): open, under investigation.
- Receipt R-04 (#106,953,944): open, under investigation.
- Receipt R-05 (#106,954,035): open, under investigation.

- Receipt R-06 (#106,954,058): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **BLS Aggregate Signatures for Signer Lists** (custody) Add an optional sfMPCPublicKey (STBlob, BLS12-381 compressed point) to the SignerList object, settable via a new tfMPCMode flag on SignerListSet in pl
- **On-Ledger NFT Rental Lease Object** (gaming) Add a new owner-reserved ledger object NFTokenLease (ltNFTOKEN_LEASE, keylet::nftLease(nftID)) storing sfNFTokenID, sfLessee, sfLeaseExpiration (ledge
- **AMM-Seeded Exit Liquidity for Fractional NFTs** (nft) Extend g12's NFTokenFractionalize transaction with a tfSeedAMM flag: after locking the NFToken into the NFTokenVault and minting the fixed-supply MPTo

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-13 04:28	09-13 04:28	xrpl-validator	`e094350`	fix(xrpl-ledger): F267 — brokered NFT accept must cover the ask after the broker fee (tecINSUFFICIENT_PAYMENT)
09-13 12:22	09-13 12:43	xrpl-validator	`123e9a7`	fix(xrpl-ledger): F268 — an out-limited leg-B book partial on the XRP bridge ceils the mid-leg strictly
09-13 12:42	09-13 12:43	xrpl-validator	`9f9f231`	fix(xrpl-ledger): F270 — a book leg enters the single-strand quality function fee-free
09-13 12:42	09-13 12:43	xrpl-validator	`1eb4a78`	fix(xrpl-ledger): F269 — a taker frozen on its own account cannot cross; the strand is never built
09-13 12:49	09-13 13:57	xrpl-validator	`c748500`	fix(xrpl-ledger): F271 — the reverse-pass sizing ladder escalates past a rung that yields nothing
09-13 13:04	09-13 13:57	xrpl-validator	`a7c12f5`	fix(xrpl-ledger): F272 — an in-limited forward pass floors the bridged mid-leg; zero drops is a dry bridge
09-13 13:13	09-13 13:57	xrpl-validator	`617c445`	fix(xrpl-ledger): F273 — the offer walk counts each pool slice as a flow iteration and stops at thirty
09-13 15:36	09-13 15:56	xrpl-validator	`d48c66a`	fix(xrpl-ledger): F274 — a FillOrKill sell is judged on rippled's per-iteration sixteen-digit fold
09-13 15:50	09-13 15:56	xrpl-validator	`88794e2`	fix(xrpl-ledger): F275 — a "became unfunded" reap lives in its trial's sandbox and dies with a rejected trial
09-13 21:48	—	xrpl-validator	`03b96c8`	fix(xrpl-ledger): F276 — an offer rate whose quotient reaches 2^63 wraps through Number's signed mantissa
09-13 22:14	—	xrpl-validator	`5d804cc`	fix(xrpl-ledger): F277 — a "became tiny" or dust reap is the strand's own; only a found-tiny one is permanent

Times in UTC. A commit appears on the day it was committed to main or merged into it.